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# Inferring the Magnetic Field from the Rayleigh–Taylor Instability

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## Abstract

By extending the potential approach of Layzer and Goncharov to the magnetohydrodynamics equations, we find the nonlinear solutions to the single-mode Rayleigh–Taylor instability subjected to uniform magnetic fields at various inclinations. This allows us to derive the analytical prediction of the terminal bubble and spike velocities at arbitrary Atwood numbers, which are assessed against various 2D and 3D direct numerical simulations. Contrary to the linear phase, where the magnetic field inhibits or delays the instability, the growth rate may be enhanced in the nonlinear regime, exhibiting velocities faster than the Alfvén speed. This sheds light on the importance of the nondimensional number expressing the competition between the magnetic and buoyancy effects. Conversely, we show how the orientation and the intensity of the magnetic field can be simply inferred from these solutions.

*Unified Astronomy Thesaurus concepts:* Magnetohydrodynamics (1964); Magnetohydrodynamical simulations (1966)

## 1. Introduction

The Rayleigh–Taylor instability (RTI; Rayleigh 1882; Taylor 1950), occurring when a light fluid is accelerated into a heavier one, is a classical mechanism explaining how variable density materials mix together. It plays a crucial role in many astrophysical or geophysical phenomena, such as supernova explosions (Hester et al. 1996; Cook & Cabot 2006; Stone & Gardiner 2007a; Porth et al. 2014; Kuranz et al. 2018), prominences in the solar corona (Isobe et al. 2005; Chae 2010; Ryutova et al. 2010; Hillier et al. 2012; Keppens et al. 2015; Terradas et al. 2015; Hillier 2018; Mishra et al. 2018; Jenkins & Keppens 2022; Changmai et al. 2023), or plasma bubbles in the ionosphere (Ott 1978; Sultan 1996; Wu 2017). The RTI is also a subject of importance for numerous engineering applications, such as in inertial confinement fusion (Betti & Hurricane 2016; Remington et al. 2019; Rigon et al. 2021; Walsh 2022). A large variety of regimes are exhibited during the RTI process (Sharp 1984; Boffetta & Mazzino 2017; Zhou 2017; Zhou et al. 2021). The linear development of the interface is quickly followed by a potential nonlinear evolution of a layer composed of rising bubbles and falling spikes (Cook & Dimotakis 2001). The transition to turbulence sees the formation of a mixing zone whose dynamics eventually converge to a self-similar phase (Dimonte et al. 2004; Ristorcelli & Clark 2004; Gréa 2013; Youngs 2013; Soulard et al. 2015). One fascinating feature of the RTI is its ability to efficiently convert potential energy into mixing (Davies Wykes & Dalziel 2014).

Beyond the purely hydrodynamic configuration, a crucial element in astrophysical or atmospherical plasmas is the presence of ambient magnetic fields. In the solar corona, for instance, the magnetic tension is responsible for the violent solar eruptions (Amari et al. 2014) or the levitation of prominences (Hillier et al. 2011; Hillier 2018). During the linear phase of the RTI, a uniform magnetic field may damp or stabilize the Rayleigh–Taylor growth (Kruskal & Schwarzschild 1954; Chandrasekhar 1961; Hillier 2016; Vickers et al. 2020). By contrast, RTI simulations with a uniform magnetic field show a drastic enhancement of the layer velocity in the nonlinear regime (Jun et al. 1995; Stone & Gardiner 2007b). The imprint of a vertical magnetic field in RTI is also revealed by the very characteristic elongated density structures. Besides, during the late-time turbulent regime, the self-similar growth is attenuated owing to an additional magnetic dissipation as recently discussed in Briard et al. (2022).

In this work, we wish to pursue the investigation further and disentangle the complex RTI dynamics driven by a uniform magnetic field. The strong coupling between the magnetic field and the RTI dynamics suggests that the bubble or spike velocity measurements in an RTI layer can be used to infer the intensity of the magnetic field. This idea has been previously considered to evaluate, for instance, the intensity of the magnetic field in a quiescent solar prominence (Berger et al. 2010; Ryutova et al. 2010; Carlyle et al. 2014; Mishra et al. 2018) or supernovae (Hester et al. 1996; Porth et al. 2014). However, due to the lack of a sound theory, it is often crudely assumed that the layer growth corresponds to the linear regime and that the magnetic field orientation is fully known. This is generally not the case for well-developed bubbles or spikes observed in astrophysical objects. Can this approach be extended to the nonlinear regime and to an arbitrary oriented magnetic field?

The potential solutions derived at the tips of bubbles or spikes (Layzer 1955; Goncharov 2002) have been for a long time the cornerstone of our understanding of the RTI dynamics. These solutions allow us to compute the so-called terminal velocity reached by bubbles or spikes. It also nicely explains why large bubbles or spikes move faster compared to smaller ones even though they have a smaller linear growth. Noticeably, the potential model equations can recover the linear evolution of the interface or be used to construct buoyancy drag and bubble merger models accounting for the turbulence regime (Ramaprabhu et al. 2006). These approaches were also successfully extended to ablative RTI configurations (Betti & Sanz 2006), to two-dimensional RTI with a horizontal

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magnetic field perpendicular to the single-mode perturbation (Gupta et al. 2010), and more recently to the context of a partially ionized plasma (Cauvet et al. 2022). The potential models have some limitations (Mikaelian 2008), and in particular they cannot account for the bubble/spike reacceleration when the secondary shear instabilities develop (Ramaprabhu et al. 2006; Wei & Livescu 2012; Bian et al. 2020). However, nonpotential extensions have been proposed that can have promising applications (Betti & Sanz 2006).

This work is organized as follows: We first present the basics equations defining the theoretical framework of this study. Then, we revisit the potential approach in order to derive the terminal bubble and spike velocities in the presence of a uniform vertical magnetic field. The last section aims at generalizing this approach to an oblique magnetic field.

## 2. Basic Equations

We consider the magnetohydrodynamics equations for two incompressible, nonviscous fluids of uniform densities $\rho^{(1)}$ and $\rho^{(2)}$ ($\rho^{(1)}>\rho^{(2)}$). The velocity $\boldsymbol{u}$, the pressure $p$, and the magnetic fields $\boldsymbol{B}$ are expressed in a Cartesian frame of coordinates $x$, $y$, $z$ and depend also on the time $t$. The vertical direction is along $z$ ($x$, $y$ thus defining the horizontal plane) with the constant gravity vector $\boldsymbol{g}$ oriented toward $z<0$, and $\mu_0$ is the permeability of free space. In this context, the incompressible velocity $\boldsymbol{u}$ and the magnetic $\boldsymbol{B}$ field verify the momentum and induction equations as

$$\rho^{(1,2)}(\partial_t\boldsymbol{u}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}) = -\nabla\left(p+\frac{\boldsymbol{B}\cdot\boldsymbol{B}}{2\mu_0}\right) + \frac{1}{\mu_0}\boldsymbol{B}\cdot\nabla\boldsymbol{B}+\rho^{(1,2)}\boldsymbol{g}, \tag{1}$$

$$\partial_t\boldsymbol{B}+\boldsymbol{u}\cdot\nabla\boldsymbol{B} = \boldsymbol{B}\cdot\nabla\boldsymbol{u}-\nabla\times\left(\frac{(\nabla\times\boldsymbol{B})\times\boldsymbol{B}}{\mu_0 n_e}\right)+\mathcal{D}\Delta\boldsymbol{B}, \tag{2}$$

$$\nabla\cdot\boldsymbol{B}=\nabla\cdot\boldsymbol{u}=0. \tag{3}$$

On the right-hand side of Equation (1), the Lorentz force expressing the effect of the magnetic field on the velocity is already decomposed into the magnetic pressure and magnetic tension terms. Here we consider a general equation for the magnetic field, Equation (2), accounting for the Hall effect ($n_e$ being the electron density) and the magnetic diffusivity expressed by its coefficient $\mathcal{D}$ (see, e.g., Galtier 2016). As will be detailed later, these later terms do not, however, play an active role in the solutions.

The two fluids (the heavy one (1) on the top of the lighter one (2)) are separated by an interface determined by the equation $z=\eta(x,\ y,\ t)$. The two kinematic conditions at the interface can be written as

$$\partial_t\eta+u_x^{(1,2)}\partial_x\eta+u_y^{(1,2)}\partial_y\eta=u_z^{(1,2)}, \quad \text{at } z=\eta(x,y,t). \tag{4}$$

In Equation (4) and whenever it is necessary in this work, the superscript $^{(1,2)}$ indicates which fluid quantity is considered.

In addition, we also write the normal stress balance expressing the dynamical condition at the interface. For nonviscous fluids and from Equation (1), the components of the stress tensor $\boldsymbol{\tau}^{(1,2)}$ for both fluids are defined as

$$\tau_{ij}=\frac{1}{\mu_0}B_iB_j-\left(p+\frac{1}{2}\frac{B_lB_l}{\mu_0}\right)\delta_{ij}, \tag{5}$$

where the summation rule is used on repeated indices. The jump condition expressing the normal stress balance is then given by

$$(\tau_{ij}^{(1)}-\tau_{ij}^{(2)})n_j=0 \text{ at } z=\eta(x,y,t), \tag{6}$$

with $\boldsymbol{n}$ the normal vector oriented toward the heavy fluid (1).

## 3. The Vertical Magnetic Field Configuration

In this section, we investigate the case of a bubble or spike propagating in a fluid at rest with the presence of a background vertical magnetic field $\boldsymbol{B}^0=(0,\ 0,\ B^0)$ (normal to the initial interface). The analysis is greatly simplified, as the magnetic field, the gravity, and the bubble/spike propagation directions are confounded.

### 3.1. The Bubble/Spike Terminal Velocity

Following Layzer (1955) and Goncharov (2002), we search for a solution of Equations (1)–(3) around the tip of an ascending bubble at $x=y=0$, $z=\eta$ and verifying the jump conditions given by Equations (4) and (6) (see also Figure 1(a)). Developing the symmetrical interface function $\eta$ at second order in $x$ and $y$, we can write

$$\eta(x,y,t)=V_zt+\mathcal{C}(x^2+sy^2), \tag{7}$$

where it is convenient to introduce an integer $s\in\{0;1\}$ for a 2D ($s=0$) or 3D geometry ($s=1$). Equation (7) expresses the rise of a bubble along the $z$-axis at constant velocity $\boldsymbol{V}=(0,\ 0,\ V_z)$ and curvature $\mathcal{C}$ to be determined.

It is now assumed that the velocity for both fluids derives from a potential function such that $\boldsymbol{u}^{(1,2)}=\nabla\phi^{(1,2)}$. We express the potential $\phi$ as

$$\phi^{(1)}(x,y,z,t)=(\mathcal{I}_x^{(1)}\cos kx+s\mathcal{I}_y^{(1)}\cos ky) \times\exp[-k(z-V_zt)] \text{ for } z\geqslant\eta, \tag{8}$$

$$\phi^{(2)}(x,y,z,t)=(\mathcal{I}_x^{(2)}\cos kx+s\mathcal{I}_y^{(2)}\cos ky) \times\exp[k(z-V_zt)]+\mathcal{K}^{(2)}(z-V_zt) \text{ for } z\leqslant\eta, \tag{9}$$

introducing $k>0$ where $k$ is the horizontal wavenumber. By construction, these solutions verify the incompressibility condition Equation (3) as $\Delta\phi^{(1,2)}=0$.

We now inject the potential functions given by Equations (8) and (9) into the kinematic conditions given by Equation (4). Developing at leading order around the tip of the bubble $x=y=0$, $z=\eta$ and using Equation (7), we recover the relations already derived in Goncharov (2002):

$$\mathcal{I}_x^{(1)}=-\frac{V_z}{k},\ \mathcal{C}=-\frac{k}{6},\ \mathcal{I}_x^{(2)}=0\ \ \mathcal{K}^{(2)}=V_z \text{ for } s=0 \text{ (2D)}, \tag{10}$$

$$\mathcal{I}_x^{(1)}=\mathcal{I}_y^{(1)}=-\frac{V_z}{k},\ \mathcal{C}=-\frac{k}{8},\ \mathcal{I}_x^{(2)}=I_y^{(2)}=0 \quad \mathcal{K}^{(2)}=V_z \text{ for } s=1 \text{ (3D)}. \tag{11}$$

Only the curvature $\mathcal{C}$ differs in the 2D /3D solutions provided by Equations (10) and (11).

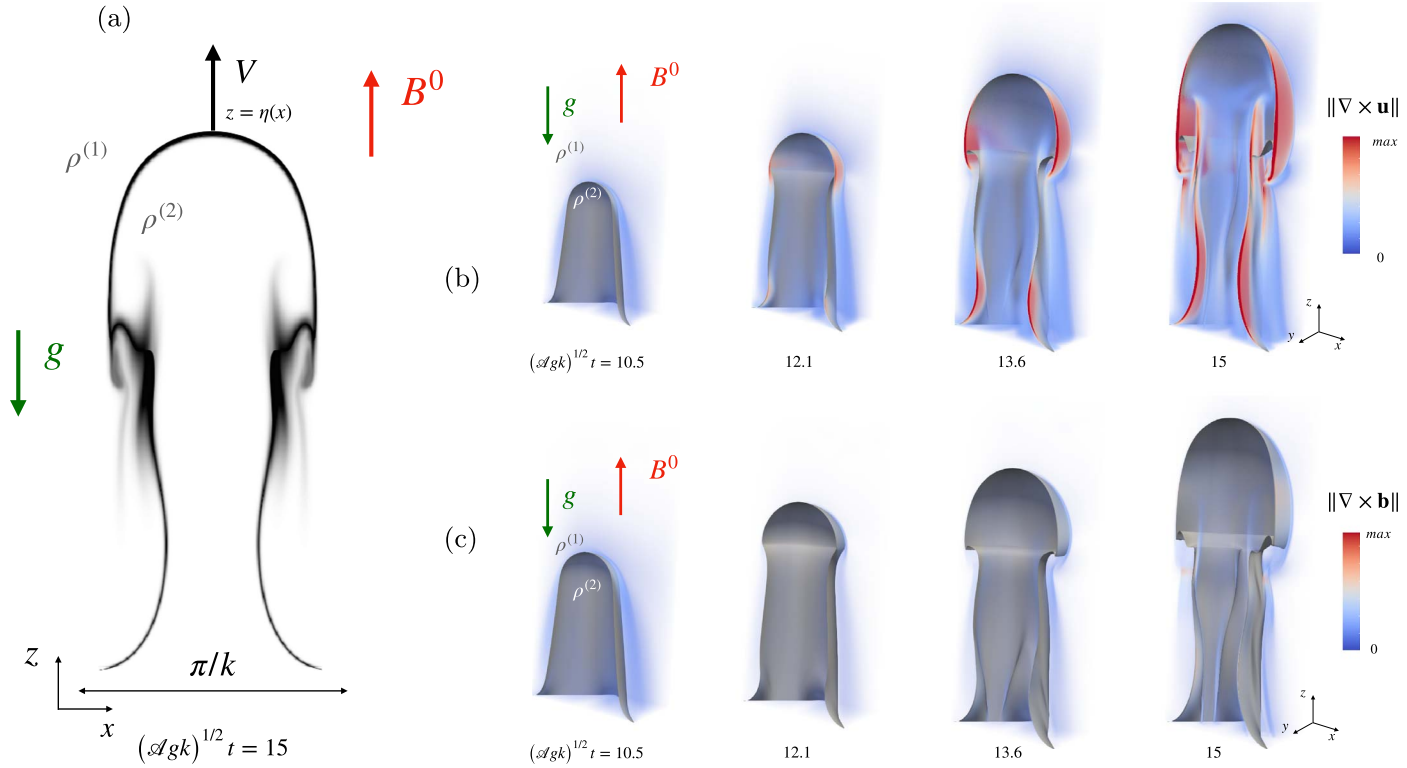



**Figure 1.** (a) Ascending Rayleigh–Taylor bubble driven by buoyancy forces and propagating in a vertical magnetic field $B^0$. The image showing the interface is extracted from a 3D simulation (S3Da; see Table 1 in the Appendix) with $g = 3$, $\mathcal{A} = 0.05$, $k = 4$, $B^0 = 0.2$, and $t(\mathcal{A}gk)^{1/2} = 15$, corresponding to the constant-velocity regime. The modulus of (b) the vorticity and (c) the curl of the magnetic field are plotted at different times in volume rendering for the same simulation. The gray isocontour delimits the interface between the heavy and the light fluids.

At this stage, and before invoking the normal stress condition at the interface, it is necessary to find a solution for the magnetic field. As the magnetic field away from the bubble is assumed to be vertical and of intensity $B^0$, it is convenient to introduce the magnetic field fluctuation $\boldsymbol{b}$ such that

$$\boldsymbol{B} = \boldsymbol{B}^0 + \boldsymbol{b}. \tag{12}$$

In this context, a potential solution for the induction equation, Equation (2), can be directly found by posing

$$\boldsymbol{b}^{(1,2)} = -\frac{B^0}{V_z}\boldsymbol{u}^{(1,2)}. \tag{13}$$

Therefore, the solution for $\boldsymbol{B}$ is irrotational. Noticeably, the terms corresponding to the magnetic diffusivity and Hall's effect are zero in Equation (2). Besides, it corresponds to a so-called force-free magnetic field, as the Lorentz force vanishes in the pure fluids. As a consequence, the Bernoulli equation for the pressure applies in each fluid, giving

$$p^{(1,2)} = f^{(1,2)}(t) - \rho^{(1,2)}\left(\partial_t \phi^{(1,2)} + \frac{1}{2}u_i u_i + gz\right), \tag{14}$$

where, due to the spatial integration, the time functions $f^{(1,2)}$ are introduced.

Using the solutions for the velocity and magnetic fields (Equations (13) and (10)) and the expression for the pressure (Equation (14)), it is possible to express the normal stress balance at the interface (Equation (6)). Similarly to Goncharov (2002), we develop at leading order around $x = y = 0$, $z = \eta$, and one obtains after some lengthy algebra the solution for the bubble velocity

$$\boldsymbol{V} = \begin{pmatrix} V_x = 0 \\ V_y = 0 \\ V_z = V \end{pmatrix} \text{with } V$$
$$= \begin{cases} \sqrt{\frac{Ag}{k(1+\mathcal{A})}\left(\frac{2}{3} + F\right)}, \text{ for } s = 0 \text{ (2D)}, \\ \sqrt{\frac{Ag}{k(1+\mathcal{A})}(2 + F)}, \text{ for } s = 1 \text{ (3D)}, \end{cases} \tag{15}$$

where the magnetic parameter $F = \frac{k}{Ag}\frac{(B^0)^2}{\rho_0 \mu_0}$ (with $\rho_0 = (\rho_1 + \rho_2)/2$) is introduced to express the balance between the magnetic and buoyancy effects. The nondimensional number $F$ can indeed be interpreted as the square of the ratio between the Alfvén frequency and the Rayleigh–Taylor inviscid growth rate for a wavenumber $k$. One can remark that setting $F = 0$ (or $B^0 = 0$) leads to the hydrodynamics solutions already derived in Goncharov (2002).

Therefore, Equation (15) indicates that the bubble velocity is enhanced in the presence of a vertical magnetic field. When increasing the magnetic parameter $F$, the bubble moves faster than the hydrodynamic case, reaching asymptotically the Alfvén velocity associated with the heavy fluid (while remaining larger than it). This is consistent with a potential solution for the magnetic fluctuation $\boldsymbol{b}$, as no Alfvén waves can escape from the interface and propagate into the heavy fluid. Also from Equation (15), while the 2D bubbles are slower than the 3D ones, the difference is somewhat reduced when the magnetic field is strong.

Following Goncharov (2002), the spike velocities can be deduced from the same procedure or by simply inverting the sign of the gravity $g$ and the Atwood number $\mathcal{A}$ in Equation (15).

### 3.2. Comparisons with the Direct Numerical Simulations

In order to assess the validity of the theoretical predictions developed in Section 3.1, we perform various 2D and 3D direct numerical simulations (DNS) with our pseudospectral code Stratospec (Gréa & Briard 2019; Viciconte et al. 2019; Briard et al. 2020, 2022). This code solves the MHD equations under the Boussinesq approximation, accounting also for viscosity, scalar, and magnetic diffusivities but discarding Hall's effect. Therefore, the comparisons are limited here to small density contrasts, although the theory is general. The simulations are performed in an elongated domain of size $2\pi \times 2\pi \times 4\pi$ using $1024^2 \times 2048$ grid points (for the 3D configurations). The details concerning the simulations, in particular their initialization and the full list of the parameters considered, are provided in the Appendix.

Figures 1(b) and (c) present the typical evolution of the interface in the simulations with a vertical magnetic field. It clearly reveals that the vorticity is well localized at the interface, confirming the validity of the potential hypothesis. In addition, the magnetic field is irrotational as expected in the pure fluid regions. This illustrates the nonlinear potential regimes and the reacceleration phase driven by the secondary shear instabilities. One of the striking manifestations of the magnetic field is the delayed apparition of vortices at the interface, leading to very stretched bubbles and spikes, as also observed in Briard et al. (2022) for 3D multimode DNS. This is a direct illustration of how a sheared interface is stabilized by a magnetic field parallel to it (see Chandrasekhar 1961).

In order to quantify the effect of $B^0$, Figures 2(a)–(d) show the time evolution of the 2D and 3D bubble heights $h_b$ and their velocities $\dot{h}_b$. Consistently with Chandrasekhar (1961), we recover that the RTI growth rate in the linear phase is attenuated by the presence of a vertical magnetic field $B^0$. This trend is inverted when entering the potential nonlinear regime as indicated in Section 3.1. More particularly, the validity of Equation (15) is assessed, as the plateau for $\dot{h}_b$ corresponds indeed to the theoretical prediction $V_z$. We remark that the duration of the plateau is larger when the magnetic field is intense, as it stabilizes the secondary Kelvin–Helmholtz instabilities. Interestingly, the terminal velocity regime occurs at larger amplitude $kh_b$ when the magnetic field becomes stronger, indicating the presence of more anisotropic structures.

In Figure 3, the renormalized terminal bubble velocities (also referred to as the Froude number in various references; Ramaprabhu et al. 2006) for the 2D and 3D simulations are reported as a function of the magnetic parameter $F$. This confirms the very good predictions provided by the potential theory.

## 4. The Oblique Magnetic Field Configuration

In this section, we extend the previous potential theory to predict the terminal bubble/spike velocities in the context of a magnetic field inclined by an angle $\theta$ with respect to the vertical (see Figure 4). The magnetic field expressed in the referential $(x, y, z)$ is thus of the form $\boldsymbol{B}^0 = (B^0 \sin\theta, 0, B^0 \cos\theta)$. The first subsection is dedicated to the derivation of the 2D and 3D potential solutions. Then, we discuss the validity of the solutions from the DNS results.

### 4.1. The Potential Regime Solutions

We start again the analysis of the bubble configuration. To this aim, it is convenient to introduce a new frame where the bubble interface is locally symmetric. This particular frame $(X, Y, Z)$ is obtained by the rotation of an angle $\alpha$ (to be determined) of the initial referential $(x, y, z)$ around the $y$-axis. In this new frame, the interface is described by the equation

$$\eta(X, Y, t) = V_Z t + \mathcal{C}(X - V_X t)^2 + s\mathcal{C}Y^2, \tag{16}$$

with again $s \in \{0; 1\}$ expressing a 2D or 3D configuration. In Equation (16), the bubble rises at a constant velocity $\boldsymbol{V} = (V_X, 0, V_Z)$, not necessarily aligned with the bubble normal, at $X = V_X t$, $Y = 0$, $Z = \eta$. The parameter $\alpha$ can be interpreted as the incidence angle of the bubble. As we will see, both configurations $\alpha \geqslant \theta$ and $\alpha \leqslant \theta$ can be encountered in the simulations. The magnetic field components in the new frame are thus provided by $B^0_Z = B^0 \cos(\theta - \alpha)$ and $B^0_X = B^0 \sin(\theta - \alpha)$ (see again Figure 4).

We then seek a potential solution for the velocity, $\boldsymbol{u} = \nabla\phi$, with

$$\begin{aligned}\phi^{(1)}(X, Y, Z, t) = (\mathcal{I}^{(1)}_X \cos[K(X - V_X t)] + s\mathcal{I}^{(1)}_Y \cos[KY]) \\ + \mathcal{J}^{(1)} \sin[K(X - V_X t)] \exp[-K(Z - V_Z t)] \text{ for } Z \geqslant \eta,\end{aligned} \tag{17}$$

$$\begin{aligned}\phi^{(2)}(X, Y, Z, t) = (\mathcal{I}^{(2)}_X \cos[K(X - V_X t)] + s\mathcal{I}^{(2)}_Y \cos[KY]) \\ + \mathcal{J}^{(2)} \sin[K(X - V_X t)] \exp[K(Z - V_Z t)] + \mathcal{K}^{(2)}(Z - V_Z t) \text{ for } Z \leqslant \eta,\end{aligned} \tag{18}$$

where the wavenumber $K$ in the rotated frame is provided by $K = k/\cos\alpha$. The magnetic field is also decomposed into a background and a perturbation part as in Equation (12). Injecting the expressions for $\phi$, Equations (17)–(18), into the equation for the induction, Equation (2), one eventually finds a potential solution for $\boldsymbol{b}$ and a condition for the velocity $\boldsymbol{V}$ as

$$\boldsymbol{b}^{(1,2)} = -\frac{B^0_Z}{V_Z}\boldsymbol{u}^{(1,2)} \text{ and} \frac{B^0_Z}{V_Z} = \frac{B^0_X}{V_X}. \tag{19}$$

The last condition in Equation (19) indicates that the bubble moves along the background magnetic field direction, $\boldsymbol{B}^0 \| \boldsymbol{V}$.

By imposing the kinematic conditions given by Equation (4) on the potential solution and expanding around $X = V_X t$, $Y = 0$, and $Z = \eta$, we obtain

$$\begin{aligned}&\mathcal{I}^{(1)}_X = -\frac{V_Z}{K},\ \mathcal{J}^{(1)} = -\frac{V_X}{2K},\ \mathcal{C} = -\frac{K}{6},\ \mathcal{I}^{(2)} = 0,\\ &\mathcal{J}^{(1)} = \frac{V_X}{4K},\ \mathcal{K}^{(2)} = V_Z, \text{ for } s = 0 \text{ (2D)},\end{aligned} \tag{20}$$

$$\begin{aligned}&\mathcal{I}^{(1)}_X = \mathcal{I}^{(1)}_Y = -\frac{V_Z}{K},\ \mathcal{J}^{(1)} = -\frac{V_X}{3K},\ \mathcal{C} = -\frac{K}{8},\\ &\mathcal{I}^{(2)}_X = \mathcal{I}^{(2)}_Y = 0,\ \mathcal{J}^{(1)} = \frac{V_X}{5K},\ \mathcal{K}^{(2)} = V_Z, \text{ for } s = 1 \text{ (3D)}.\end{aligned} \tag{21}$$

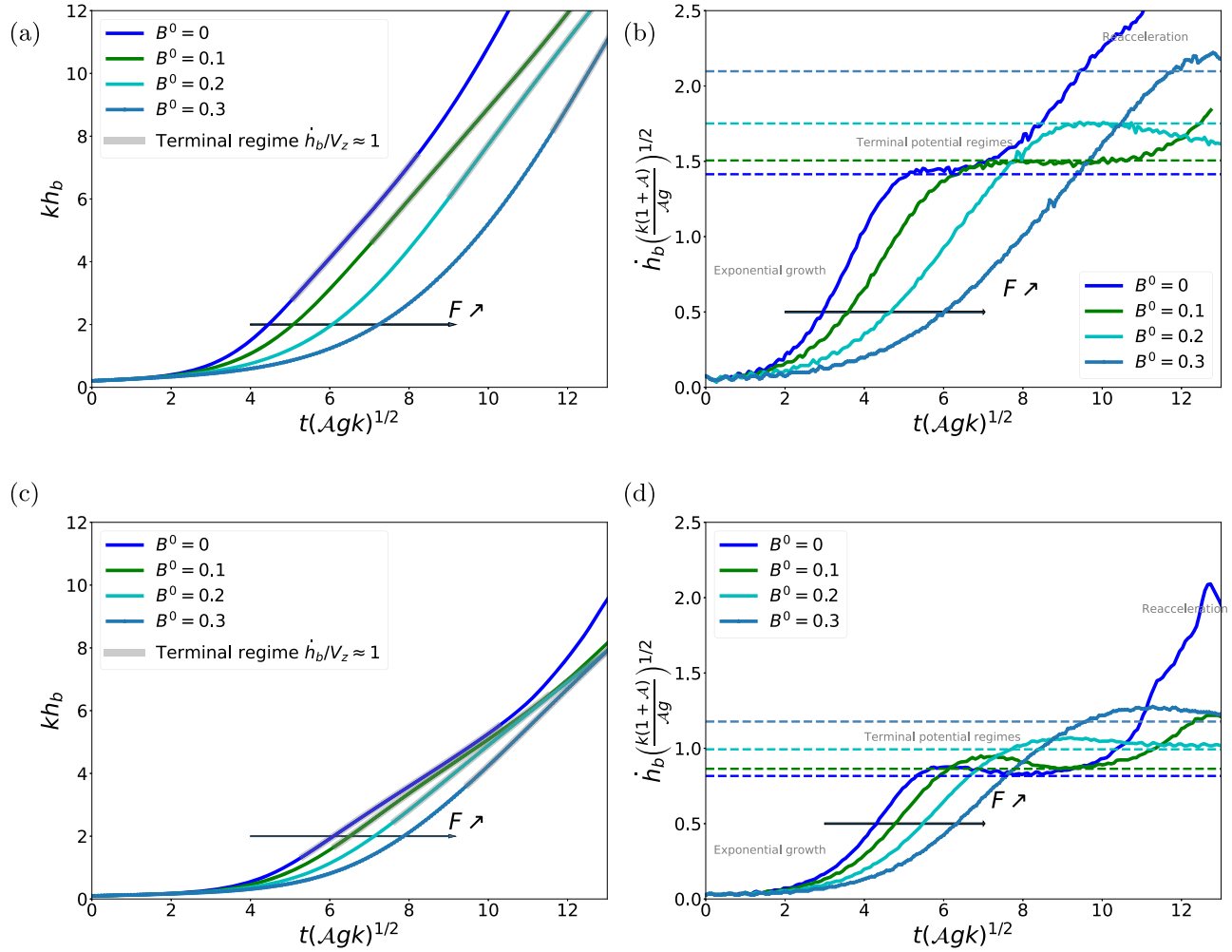



**Figure 2.** Time evolution of (a) the bubble height $h_b$ and (b) its derivative in the 2D Rayleigh–Taylor simulations at various vertical magnetic fields $B^0$. The simulations correspond to S2Da–d with $k = 2$ (see Table 1 in the Appendix). Similarly, the bubble height and its time derivative corresponding to the 3D configuration S3Da with $k = 4$ are shown in panels (c) and (d), respectively. The predictions for the terminal velocity, Equation (15), are also indicated with dashed lines in panels (b) and (d).

In Equations (20) and (21) the solutions in the rotated frame are identical to the vertical magnetic configuration ones, Equations (10) and (11), with additional terms due to the bubble lateral displacement $V_X$.

In order to derive an expression for the bubble velocity, we use the Bernoulli formula to compute the pressure and apply the normal stress condition at the interface, Equation (6), in the rotated frame ($X$, $Y$, $Z$). These lengthy but straightforward calculi are performed using a Mathematica script. The full solutions providing $\boldsymbol{V}$ and $\alpha$ are parameterized by $\theta - \alpha$ but unfortunately do not have a simple expression. Moreover, the solutions are multivaluate, i.e., at a given $\theta$ different branch solutions can be found. We focus on the solutions with $\lim_{\theta\to 0}\alpha = 0$, as it recovers the symmetric bubble already derived for the vertical magnetic field configuration. Therefore, two branches $+$ and $-$ can be obtained corresponding to $\alpha \geqslant \theta$ and $\alpha \leqslant \theta$, respectively. The results are presented in Figure 5 in the limit $\mathcal{A} \to 0$, which does not depend on the magnetic parameter $F$. This also demonstrates that the incidence angle $\alpha$ can be reasonably well approximated by the asymptotic limit $\theta \to 0$. In 2D, the two $\pm$ branches are given by $\alpha = 5\theta/3$, $\theta/3$, and for the 3D configuration $\alpha = 8\theta/5$, $2\theta/5$ similarly. We use these expressions in order to derive a more tractable formula for $\boldsymbol{V}$. Eliminating $\alpha$ in the solutions and taking the limit $\mathcal{A} \to 0$, we find the following expression for the bubble

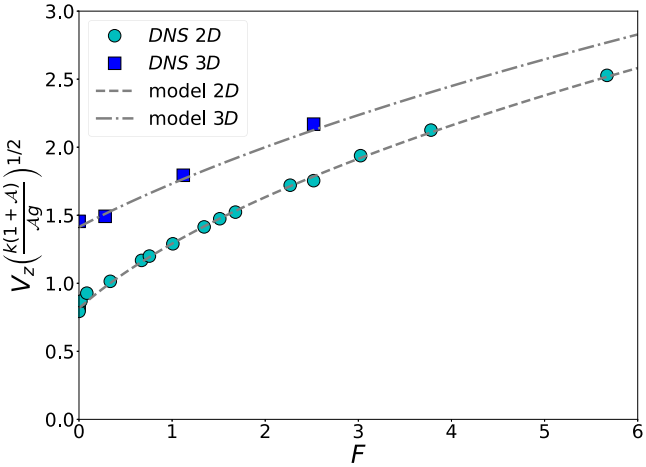



**Figure 3.** Normalized bubble velocity as a function of the parameter $F$ for the vertical magnetic field $B^0$ configuration. The symbols correspond to measurements from 2D and 3D simulations (S2Da–d and S3Da). The lines indicate the theoretical solutions given by Equation (15) derived in this work.

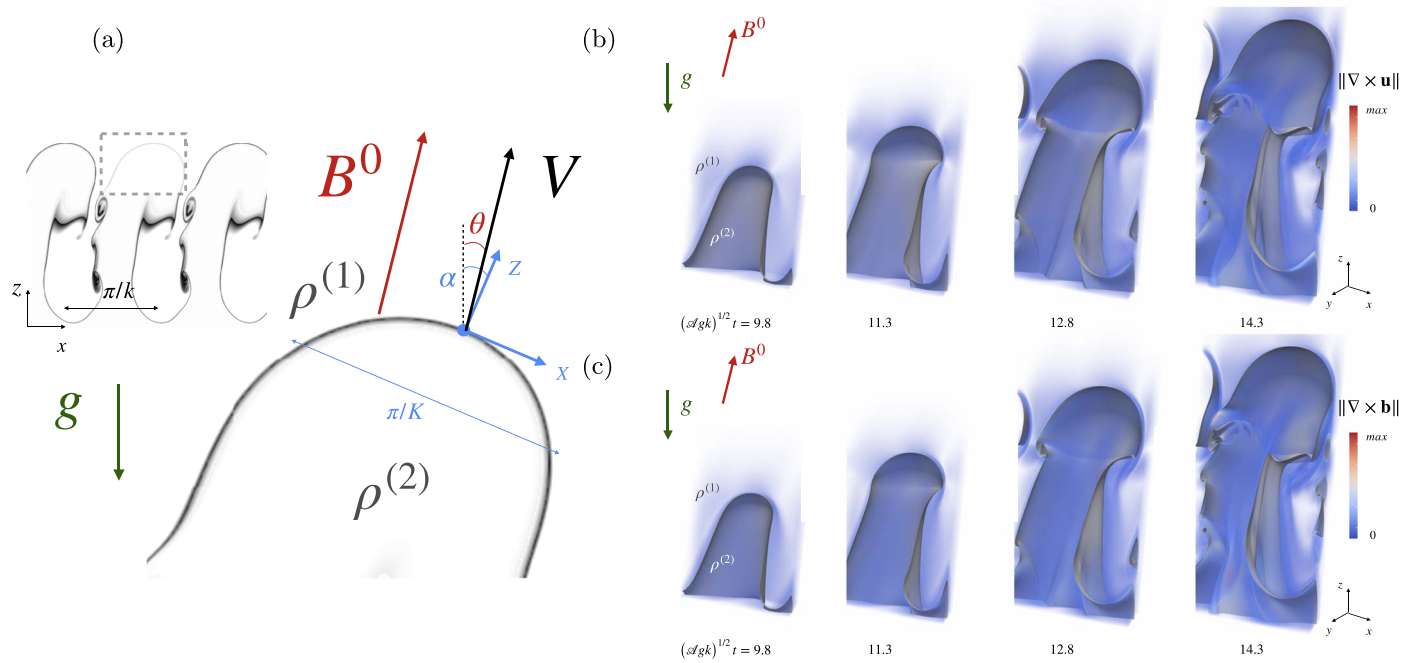



**Figure 4.** (a) Ascending Rayleigh–Taylor bubble driven by buoyancy forces and propagating in an oblique magnetic field $\boldsymbol{B}^0$. The image showing the interface is extracted from a 3D simulation (S3D; see Table 1 in the Appendix) with $g = 3$, $\mathcal{A} = 0.05$, $k = 4$, $B_z^0 = 0.2$, $B_x^0 = 0.05$, and $t(\mathcal{A}gk)^{1/2} = 15$, corresponding to the constant-velocity regime. The moduli of (b) the vorticity and (c) the curl of the magnetic field are plotted at different times in volume rendering for the same simulation. The gray isocontour delimits the interface between the heavy and light fluids.

terminal velocity:

$$V = \begin{pmatrix} V_x = V\sin\theta \\ V_y = 0 \\ V_z = V\cos\theta \end{pmatrix} \text{ with } V = \begin{cases} \sqrt{\dfrac{Ag}{K(1+\mathcal{A})}}\left(\dfrac{64\sqrt{2}}{3\sqrt{787 + 700\cos[4\theta/3] + 561\cos[8\theta/3]}} + F\right)^{1/2} & \text{for } s = 0 \text{ (2D)}, \\ \sqrt{\dfrac{Ag}{K(1+\mathcal{A})}}\left(\dfrac{900\sqrt{2}}{\sqrt{506343 - 489724\cos[6\theta/5] + 388381\cos[12\theta/5]}} + F\right)^{1/2} & \text{for } s = 1 \text{ (3D)}, \end{cases} \tag{22}$$

with $F = \dfrac{K}{Ag}\dfrac{(B^0)^2}{\rho_0\mu_0}$. Here we provide some remarks concerning the solution of Equation (22): (i) The bubble moves along the $B^0$ direction according to the potential solution (see Equation (19)) such that $V_x = \tan\theta V_z$. (ii) By setting $\theta = 0$, we recover the vertical magnetic field solutions, Equation (15). (iii) From dimensional analysis, the nondimensional vertical velocity $V_z\sqrt{K(1+\mathcal{A})/\mathcal{A}g}$ should depend on $F$ and $\mathcal{A}$. In practice, the dependence on the Atwood number is very weak, and Equation (22) stands as a good approximation even at large $\mathcal{A}$. (iv) The two branches $\pm$ are both expressed by Equation (22), as the potential solutions are symmetrical with respect to $\theta - \alpha$. However, it corresponds to different bubble orientations as $K^+ = k/\cos(5\theta/3)$, $K^- = k/\cos(\theta/3)$ in 2D and $K^+ = k/\cos(8\theta/5)$, $K^- = k/\cos(2\theta/5)$ in 3D. (v) As for the vertical configuration at large $F$, the bubble propagates at the Alfvén velocity of the heavy fluid in the direction of the background magnetic field $B^0$. (vi) By replacing $\mathcal{A} \to -\mathcal{A}$ and $g \to -g$ in Equation (22), one obtains the solution for the spikes.

In Figures 6(a) and (b), we represent the terminal bubble solution approximated by Equation (22) in polar coordinates $(F, \theta)$.

Noticeably, the magnetic fields enhance the vertical bubble velocity compared to a pure hydrodynamic configuration for a large panel of $F$ and $\theta$ values around $\theta = 0$ or $\pi$ (corresponding to the velocities above the black isoline indicated in the panels). In the region around $\theta \to \pm\pi/2$, the trend is, however, inverted. In this horizontal limit, the potential solutions give $V_z = 0$ as the velocity gets aligned with the background magnetic field $\boldsymbol{B}^0$. These solutions do not, however, apply to the usual 3D Rayleigh–Taylor observations with a horizontal magnetic field (Stone & Gardiner 2007b; Carlyle & Hillier 2017). Contrary to the assumptions made in our potential model, this configuration develops very anisotropic bubbles or spikes, as the mode in the direction of the magnetic field (i.e., $x$) is severely inhibited or delayed (and $V_x = 0$). Therefore, the dynamics would develop principally in the 2D plane $y$–$z$, and the bubble/spike terminal velocities are thus simply provided by the 2D solution of Equation (15), where $k$ is the wavenumber in the $y$-direction and $F$ is constructed from the vertical component of the magnetic field $B_z^0$.

### 4.2. Comparisons with the DNS

We now present the results from the highly resolved 2D and 3D simulations with an inclined magnetic field as detailed in Table 1 in the Appendix. These simulations exhibit a potential regime as described in Section 4.1 with a plateau reached by

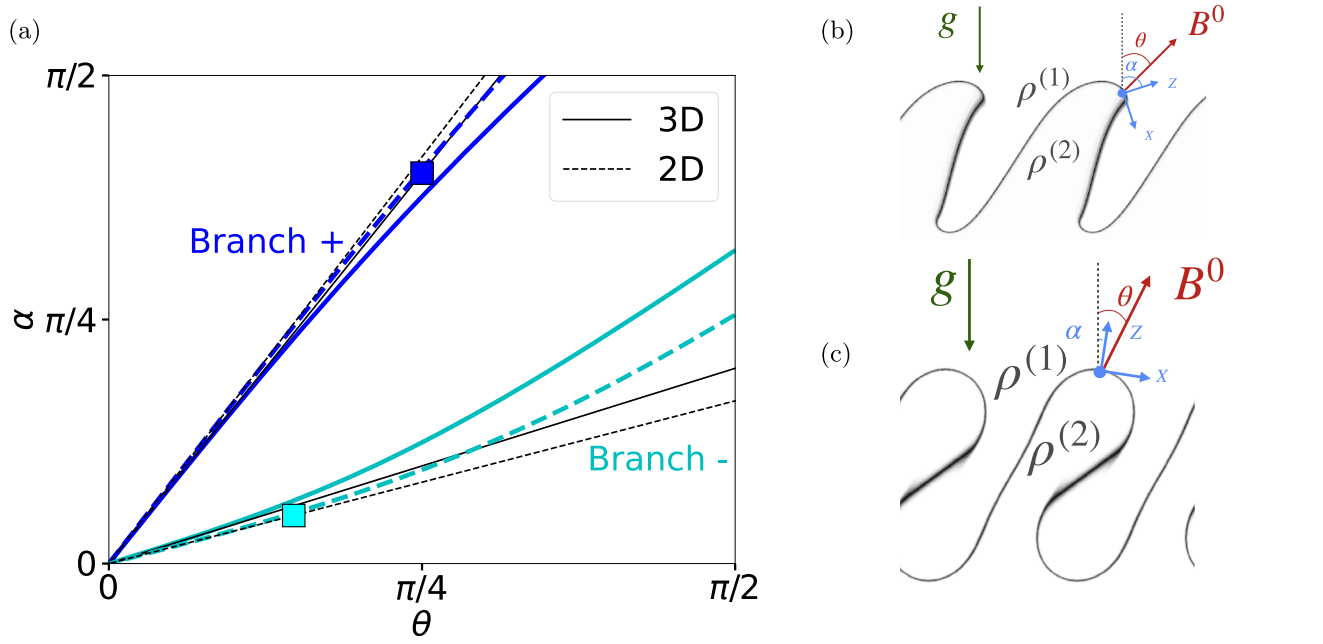



**Figure 5.** (a) The bubble incidence angle $\alpha$ as a function of the magnetic field angle $\theta$ for the two branches $\pm$ potential solutions (color curves) in the limit $\mathcal{A} \to 0$ and with 2D/3D configurations. The dashed and solid black curves indicate the asymptotic limits for these solutions at $\theta \to 0$, corresponding to $\alpha = \theta/3, 2\theta/5, 8\theta/5, 5\theta/3$ and associated with Equation (22). The right panels show the bubble interfaces extracted from (b) a 3D simulation from S3Db and (c) a 2D simulation from S2De corresponding in panel (a) to the symbols of the branches $+$ and $-$, respectively.

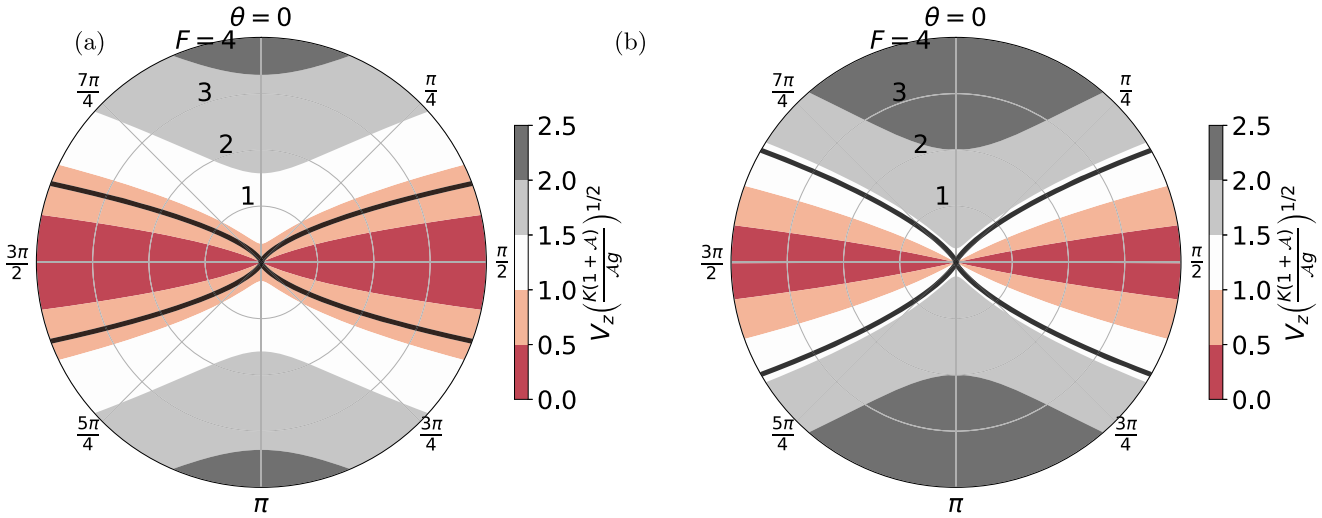



**Figure 6.** Isocontours of the renormalized terminal bubble velocity approximated by Equation (22) in a polar representation $(F, \theta)$ for (a) the 2D configuration and (b) the 3D configuration. The solid black lines correspond to the isoline associated with the velocity of the purely hydrodynamic configuration.

the bubble/spike velocity. Figures 4(b) and (c) show an example of the magnitudes of the vorticity and the curl of the magnetic field. As assumed by the theory, it is very weak in the pure fluid regions and peaked at the interface. Besides, the bubble moves along the magnetic field direction as expected.

The potential theory predicts the existence of two branch solutions associated with different incidence angles of the bubble or the spike. Figures 5(b) and (c) show that the two branch solutions are encountered in the DNS. The angle $\alpha$ can thus be recovered by considering the minimum radius of curvature along the tip of the bubble. This indeed corresponds to the location where the bubble is locally symmetric. As expected, the bubbles with a very small minimum radius of curvature are of branch $+$ type as seen in the bubble of Figure 5(b) presenting a sharp edge. Most of the 2D solutions have $\alpha \leqslant \theta$ corresponding to the branch $-$ type. However, at large $\theta$ we have observed that the branch $+$ solutions are favored in the 2D case. Conversely, all our 3D simulations are of the branch $+$ type with $\alpha \geqslant \theta$.

We gather in Figures 7(a) and (b) the values of the bubble vertical velocity when it reaches a plateau as a function of the angle $\theta$. For these DNS series, only the intensity of the horizontal magnetic field $B_x^0$ is varied. At large $\theta$, the vertical bubble velocity is reduced. The branch $+$, which corresponds to bubbles with a smaller radius of curvature and a larger wavenumber $K$ as illustrated by Figure 5(b), thus experiences a larger decrease of the velocity compared to the branch $-$. One can notice the good prediction provided by Equation (22).

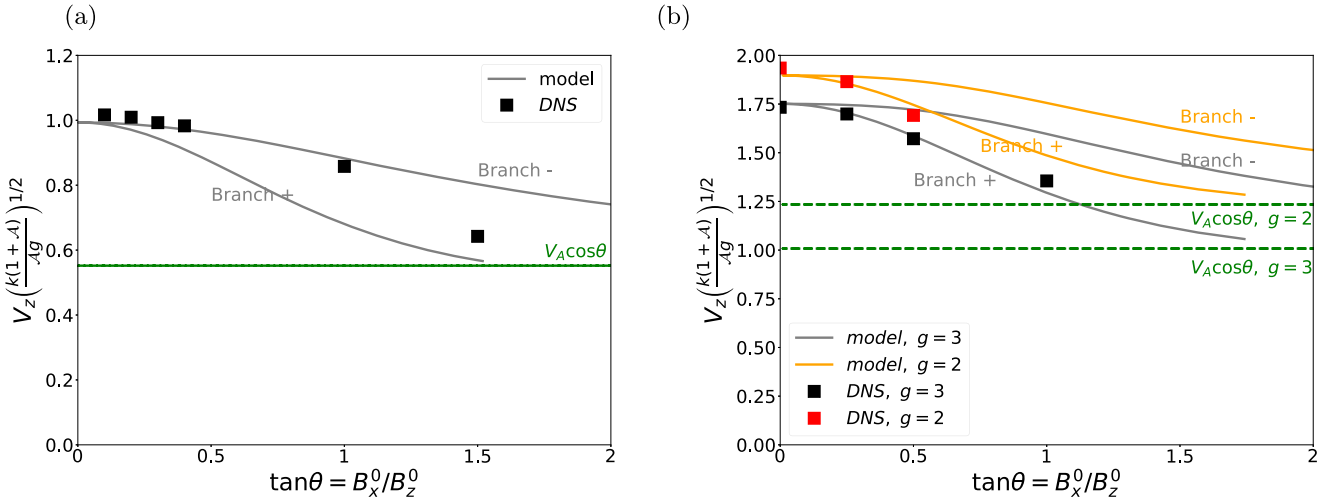



**Figure 7.** Renormalized terminal bubble velocities as a function of the orientation of the magnetic field extracted from the simulations and predicted by Equation (22). The simulation parameters are (a)$g = 5$, $\mathcal{A} = 0.05$, and $k = 2$ for the 2D configurations (S2Dd; see Table 1 in the Appendix) and (b) $g = 2$ or 3, $\mathcal{A} = 0.05$, and $k = 4$ (S3Db,c) for the 3D ones. The renormalized Alfvén velocity of the heavy fluid is provided by $V_A = B^0/\sqrt{\mu_0\rho_1}\sqrt{k(1+\mathcal{A})/\mathcal{A}g}$.

## 5. Conclusion

In this work, we derived the terminal velocities of Rayleigh–Taylor bubbles or spikes propagating in uniform magnetic fields at various directions. These nonlinear solutions are expressed for incompressible fluids with arbitrary density contrasts (or Atwood numbers) and in 2D/3D geometries, extending the classical potential theory proposed by Layzer (1955) and Goncharov (2002) to the magnetohydrodynamics accounting for a magnetic diffusivity and Hall's effect. It shows that axisymmetric bubbles/spikes move along the magnetic field lines, reaching velocities larger than the hydrodynamic case and the Alfvén velocity. The velocity magnitude is thus provided by Equation (22), exhibiting the role of the magnetic parameter expressing the competition between the magnetic and buoyancy effects. The theory also reveals that the bubble or spike inclination is not aligned with the magnetic field direction and can be predicted by two branch solutions. When the magnetic field gets aligned with the horizontal plane, bubbles or spikes are usually no longer axisymmetric, as one direction is frozen. The 2D potential solutions given by Equation (15) can be used in this limit to compute the bubble/spike terminal velocity along the vertical direction.

By the mean of highly resolved 2D and 3D single-mode Rayleigh–Taylor simulations, we assess the validity of these solutions in the limit of the Boussinesq approximation. The potential regime is reached at longer times in the MHD DNS compared to the hydrodynamic case owing to the slower linear dynamics. However, it maintains itself much longer, as the bubble or spike reacceleration is delayed by the strong magnetic damping of the secondary shear instabilities. Finally, the bubble and spike velocities measured in the simulations are well predicted by the nonlinear theory for a large panel of magnetic field orientations, giving strong confidence in its predictions. Besides, it explains why magnetic Rayleigh–Taylor layers exhibit velocities faster than the Alfvén speed when entering the turbulent regime (Stone & Gardiner 2007b; Briard et al. 2022).

These results also suggest that the magnetic field can be inferred from the Rayleigh–Taylor bubble and spike velocities measured in astrophysical objects such as solar prominences. Perhaps the simplest method consists in considering small axisymmetric bubbles or spikes, as they are expected to move at the Alfvén velocity along the magnetic lines. Besides for small objects, the hypothesis of a uniform magnetic field is more easily justified and the magnetic parameter can be computed a posteriori, knowing the bubble or spike curvature, to verify the decoupling between magnetic and buoyancy effects. If, however, the buoyancy effects are significant, it is possible to use Equation (22) to infer the magnetic field magnitude. For configurations where the magnetic field is close to the horizontal plane, bubbles and spikes are no longer axisymmetric and propagate along the vertical direction. The 2D solution given by Equation (15) comes in handy in order to derive the vertical component of the magnetic field. Hopefully, these analytical solutions will provide a useful tool and pave the way to more accurate and consistent results with the strongly nonlinear regimes observed in astrophysical Rayleigh–Taylor configurations. A path toward even more realistic solutions would be to account for the compressibility and the magnetic shear (Ruderman 2017).

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## Appendix
## Direct Numerical Simulations Using Stratospec

The pseudospectral code Stratospec solves the MHD Boussinesq Navier–Stokes equations for a binary mixture of miscible fluids in a $2\pi \times 2\pi \times 4\pi$ domain for the 3D configuration (see Briard et al. 2022). All the 3D simulations presented here have a $1024^2 \times 2048$ resolution. In order to prepare the Rayleigh–Taylor setup, we use a perturbed diffuse interface constructed from the heavy fluid mass concentration $C$ as

$$C(x, y, z, t = 0) = \frac{1}{2}(1 + \tanh[(z - \eta(x, y))/\delta]). \quad \text{(A1)}$$

In Equation (A1), the initial perturbed interface is given by $\eta = a\cos kx + sa\cos ky$, $s \in \{0; 1\}$ for a 2D or 3D configuration, the perturbation amplitude is $a = 0.01$, and the interface thickness is approximated by $3\delta$ with $\delta = 0.02$. These

**Table 1**
Parameters and Names of the DNS Series Presented in This Work

| Geometry | $\mathcal{A}$ | $g$ | $k$ | $B_z^0$ | $B_x^0$ | Name |
|---|---|---|---|---|---|---|
| 2D | 0.05 | 5 | [2; 4; 6; 8; 10; 15] | 0.3 | 0 | S2Da |
| 2D | 0.05 | 5 | [2; 4; 6; 8; 10; 15] | 0.2 | 0 | S2Db |
| 2D | 0.05 | 5 | 2 | 0.1 | 0 | S2Dc |
| 2D | 0.05 | 5 | [2; 4; 6; 8; 10; 15] | 0. | 0 | S2Dd |
| 3D | 0.05 | 3 | 4 | [0; 0.1; 0.2; 0.3] | 0 | S3Da |
| 2D | 0.05 | 5 | 2 | 0.2 | [0; 0.02; 0.04; 0.06; 0.08; 0.2; 0.3] | S2De |
| 3D | 0.05 | 3 | 4 | 0.2 | [0; 0.05; 0.1; 0.2] | S3Db |
| 3D | 0.05 | 2 | 4 | 0.2 | [0; 0.05; 0.1] | S3Dc |

**Note.** The first and last lines correspond to the vertical and inclined magnetic field configurations, respectively.

parameters allow sufficiently well resolved simulations, as around 15 points describe the interface. In addition, the velocity field is initially zero, $\boldsymbol{u}(t=0)=0$, and for the magnetic field, $\boldsymbol{B}(t=0)=\boldsymbol{B}^0$.

The kinematic viscosity, scalar, and magnetic diffusion coefficients are set to $10^{-4}$ in all the simulations. This ensures that the simulations do not diffuse too much until the end fixed at $t=18$.

The other parameters considered for the DNS are given in Table 1.

The results are taken into account as long as the size of the bubbles and spikes is less than a quarter of the domain height in order to avoid confinement due to the size of the computational domain. This also explains why we use larger $k$ for the faster 3D configurations in order to reduce the velocity in the potential regime. The 2D studies SD2a,b,d use various values of $k$ in order to show that the initial interface thickness $\delta$ and amplitude $a$, not accounted for in the potential theory, do not play an important role in the bubble/spike potential regime.

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